

GIS-Based Assessment of Vegetation Density and Built-up Surface Distribution in Jodhpur City Using Sentinel-2 Remote Sensing Data

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CHAPTER 1: INTRODUCTION

1.1 Background

Urbanization is one of the most significant land transformation processes occurring globally. Rapid population growth and infrastructure development have increased pressure on natural resources, resulting in the reduction of vegetation cover and expansion of built-up surfaces. Monitoring these changes is essential for sustainable urban planning and environmental management.

Remote sensing and Geographic Information Systems (GIS) provide effective tools for analyzing land surface characteristics over large geographic areas. Spectral indices derived from satellite imagery enable the identification and classification of vegetation and built-up land cover classes.

This study utilizes Sentinel-2 multispectral imagery to evaluate vegetation density and built-up surface distribution within Jodhpur City, Rajasthan, India. Two commonly used indices were employed:

- Normalized Difference Vegetation Index (NDVI)
- Normalized Difference Built-up Index (NDBI)

The generated thematic maps provide insights into urban growth patterns and vegetation distribution across the city.

1.2 Need for the Study

Jodhpur is experiencing continuous urban expansion due to population growth and economic development. Such growth often leads to:

- Reduction of vegetation cover
- Increased impervious surfaces
- Urban heat island effects
- Environmental degradation
- Increased demand for sustainable planning

Accurate spatial information regarding vegetation and built-up areas is therefore essential for planners and policymakers.

1.3 Study Area

Jodhpur City is located in western Rajasthan and forms an important urban center of the Thar Desert region.

Geographical Coordinates:

Latitude:

26.22°N – 26.40°N

Longitude:

72.90°E – 73.15°E

Characteristics:

- Semi-arid climate
- Low annual rainfall
- Sparse natural vegetation
- Rapid urban growth
- Rocky and barren terrain

Projection Used:

WGS 84 / UTM Zone 43N

EPSG:32643

1.4 Objectives

The major objectives of this study are:

1. To prepare a Vegetation Density Map of Jodhpur City using NDVI.
2. To prepare a Built-up Surface Density Map using NDBI.
3. To classify vegetation into different density classes.
4. To classify built-up areas into different density categories.
5. To calculate area and percentage coverage of each class.
6. To analyze the spatial distribution of vegetation and urban development.

CHAPTER 2: LITERATURE REVIEW

2.1 Remote Sensing for Urban Studies

Remote sensing has become an important tool for monitoring urban growth and environmental changes. Multispectral satellite data enable the extraction of land cover information through spectral analysis.

2.2 NDVI

NDVI is one of the most widely used vegetation indices.

Formula:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

Where:

NIR = Near Infrared Band

Red = Red Band

Higher NDVI values indicate healthier vegetation.

2.3 NDBI

NDBI is used for identifying built-up areas.

Formula:

$$\text{NDBI} = (\text{SWIR} - \text{NIR}) / (\text{SWIR} + \text{NIR})$$

Where:

SWIR = Short Wave Infrared Band

NIR = Near Infrared Band

Higher NDBI values generally indicate built-up surfaces.

CHAPTER 3: DATA AND SOFTWARE

3.1 Satellite Data

Source:

Sentinel-2 MSI

Provider:
Copernicus Open Access Hub

Spatial Resolution:
20 m

Acquisition Year:
2026

Bands Used:

For NDVI:

Band 8 (NIR)
Band 4 (Red)

For NDBI:

Band 11 (SWIR)
Band 8 (NIR)

3.2 Administrative Boundary

Jodhpur City Boundary Polygon

Used for clipping and masking analysis outputs.

3.3 Software

QGIS 3.40

Functions Used:

- Raster Calculator
- Reclassify by Table
- Raster to Polygon
- Dissolve
- Field Calculator
- Layout Manager

CHAPTER 4: METHODOLOGY

Step 1: Data Acquisition

Sentinel-2 imagery was downloaded and preprocessed.

Step 2: Image Clipping

The imagery was clipped using the Jodhpur city boundary.

Step 3: NDVI Calculation

Raster Calculator expression:

$$("B8" - "B4") / ("B8" + "B4")$$

Output:

NDVI Raster

Step 4: NDVI Reclassification

Classes Created:

Class 1:

Built-up/Bare Land

Class 2:

Sparse Vegetation

Class 3:

Moderate Vegetation

Class 4:

Dense Vegetation

Step 5: Vector Conversion

Raster to Polygon Conversion

Step 6: Dissolve Operation

Polygons belonging to the same class were merged.

Step 7: Area Calculation

Area fields were created using:

$\$area/1000000$

to obtain area in km².

Step 8: NDBI Calculation

Raster Calculator expression:

$("B11" - "B8") / ("B11" + "B8")$

Output:

NDBI Raster

Step 9: NDBI Reclassification

Classes:

Class 1:

Water / Non-built-up

Class 2:

Low Built-up

Class 3:

Moderate Built-up

Class 4:

High Built-up

Step 10: Vectorization and Dissolve

Raster converted to polygons.

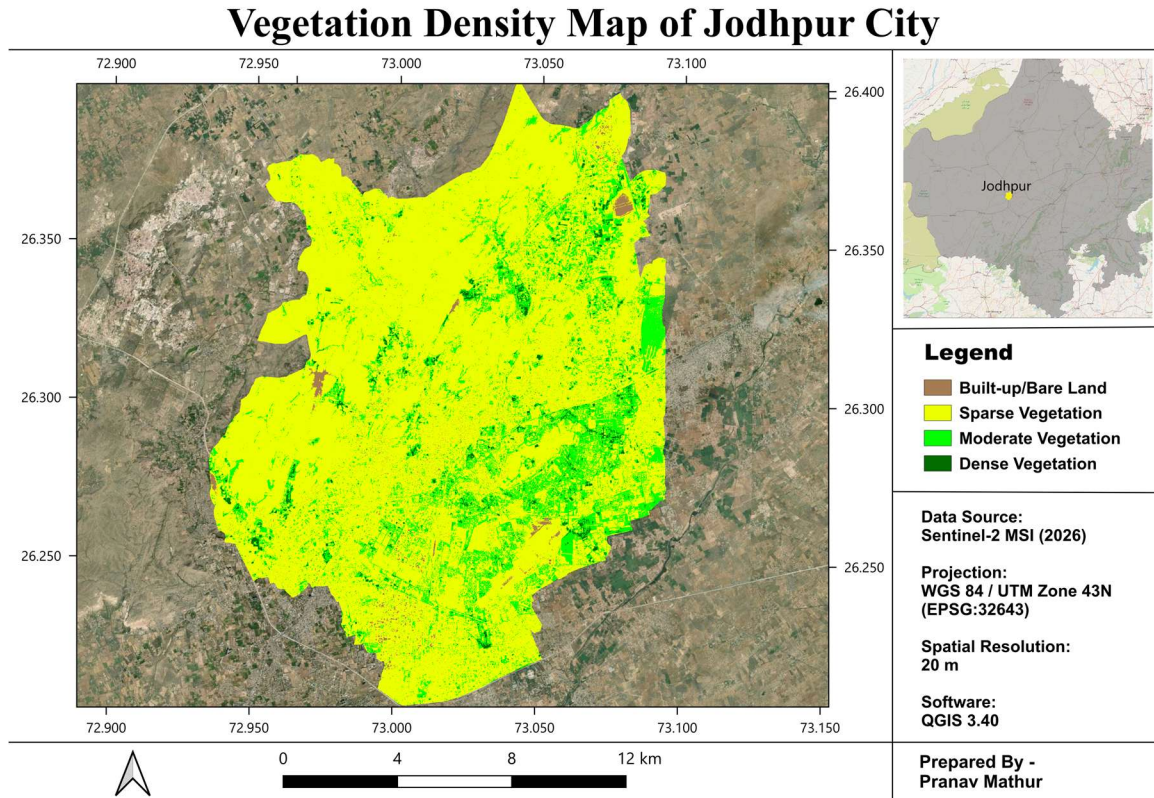
Classes dissolved based on density category.

Step 11: Area Calculation

Area and percentage calculated for each class.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Vegetation Density Analysis



The NDVI-based classification indicates that sparse vegetation dominates the study area.

Class	Area (km ²)	Percentage
Built-up/Bare Land	3	1%
Sparse Vegetation	190	81%
Moderate Vegetation	38	16%
Dense Vegetation	5	2%

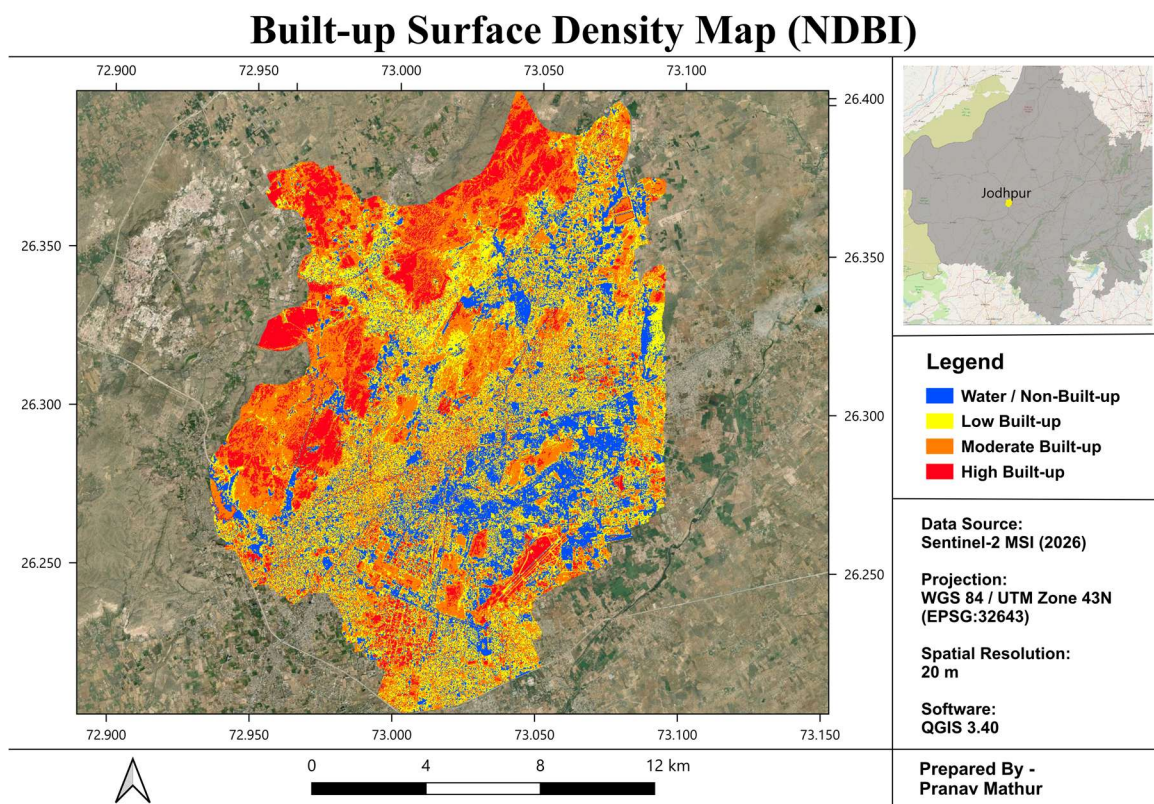
Interpretation

The predominance of sparse vegetation reflects the semi-arid nature of Jodhpur.

Dense vegetation occurs only in:

- Urban parks
- Agricultural patches
- Water-associated green zones

5.2 Built-up Surface Density Analysis



Class	Area (km ²)	Percentage
Water / Non-built-up	43	19%
Low Built-up	69	30%
Moderate Built-up	85	37%
High Built-up	31	14%

Interpretation

Moderate built-up areas occupy the largest portion of the city.

High-density urban zones are concentrated around:

- Central Jodhpur
- Commercial corridors
- Major transportation routes

5.3 Comparison of NDVI and NDBI

An inverse relationship was observed:

- Areas with high NDVI generally exhibit low NDBI.
- Areas with high NDBI generally exhibit lower vegetation density.

This confirms the expected urban–vegetation trade-off.

CHAPTER 6: LIMITATIONS

1. NDBI can misclassify barren rocky surfaces as built-up areas.
2. Mixed pixels may occur due to 20 m spatial resolution.
3. Seasonal variations affect vegetation indices.
4. Threshold values may vary across regions.
5. Atmospheric effects may introduce minor errors.

CHAPTER 7: CONCLUSION

The study successfully demonstrated the application of GIS and remote sensing techniques for urban environmental assessment in Jodhpur City. NDVI and NDBI proved effective in identifying vegetation density and built-up surface patterns. Results indicate that sparse vegetation dominates the landscape, while moderate built-up density occupies the largest urban area. The generated thematic maps can support sustainable urban planning, environmental monitoring, and future land-use management initiatives.

REFERENCES

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